

Wicara: Efforts to Reduce the Fear of Public Speaking for Glossophobia Sufferers Using a Mobile Application Prototype Based on Large Language Model

1st Dhi'fan Razaqa
Faculty of Computer Science
University of Pembangunan Nasional
"Veteran" Jakarta
Jakarta, Indonesia
2210511091@mahasiswa.upnvj.ac.id

2nd Huda Rasyad Wicaksono
Faculty of Computer Science
University of Pembangunan
Nasional "Veteran" Jakarta
Jakarta, Indonesia
2210511098@mahasiswa.upnvj.ac.id

3rd Fajar Ramadhan
Faculty of Computer Science
University of Pembangunan Nasional
"Veteran" Jakarta
Jakarta, Indonesia
2210511097@mahasiswa.upnvj.ac.id

4th Muhammad Rezka Al Maghribi
Faculty of Computer Science
University of Pembangunan Nasional
"Veteran" Jakarta
Jakarta, Indonesia
2210511086@mahasiswa.upnvj.ac.id

5th Axel Putra Bintang Syahkuala
Sebayang
Faculty of Computer Science
University of Pembangunan Nasional
"Veteran" Jakarta
Jakarta, Indonesia
2210511102@mahasiswa.upnvj.ac.id

*6th Radinal Setyadinsa
Faculty of Computer Science
University of Pembangunan Nasional
"Veteran" Jakarta
Jakarta, Indonesia
radinalsetyadinsa@upnvj.ac.id

Glossophobia, or fear of public speaking, affects 63.9% of 1,135 college students and 77.42% of 31 high school students. In one study, 50% of 95 university students reported severe anxiety, 42% moderate anxiety, and 9% low anxiety when speaking publicly. This condition stems from low self-confidence, anxiety, trembling, cold sweats, difficulty breathing, increased heart rate, and lack of preparation. The Wicara application prototype integrates artificial intelligence, specifically a large language model (LLM) to evaluate communication skills and enable users to practice public speaking. Developed using the Waterfall method, it offers features such as basic public speaking materials, three practice modes, and progress tracking. The project aims to create an innovative solution for individuals with glossophobia to reduce anxiety and improve speaking skills. Usability testing using a 7-point Likert scale USE Questionnaire yielded positive results: 6.2 for usefulness and ease of use, 6.3 for learnability, and 6.4 for satisfaction. A paired t-test statistical value 0.007 demonstrated the prototype's impact on reducing anxiety levels. These results indicate that the Wicara prototype has significant potential to benefit users in improving public speaking skills and overcoming anxiety.

Keywords—Glossophobia, Public Speaking, large language model, Wicara.

I. INTRODUCTION

Public speaking is a fundamental form of communication. Deborah Grayson Riegel, a communication practitioner, defines public speaking as the ability to convey messages clearly, influence the audience, and establish an emotional connection with listeners. Glossophobia, derived from the Greek words "glossa" (tongue) and "phobos" (fear), refers to the condition where an individual experiences fear or anxiety when required to speak publicly to convey ideas, thoughts, or opinions.

Experts estimate that approximately 75% of the population experiences varying degrees of fear related to public speaking [1]. The reality of public speaking skills among students in the Indonesian Language and Literature Education Study Program remains significantly low. On average, only 24.60% of students are active participants, while 75.40% are inactive [2]. In the study by [3], 50% of 95 students were reported to experience severe anxiety, 42% experienced moderate anxiety, and 9% experienced low anxiety during public speaking, indicating high levels of anxiety when speaking in front of an audience. Testing conducted with 52 respondents revealed that only one

individual was detected as experiencing low-level anxiety. The remaining 51 respondents were found to have moderate to very high levels of anxiety. This indicates that a significant majority 98% of the participants experienced considerable levels of anxiety, ranging from moderate to high, while only a small fraction 2% reported low levels of anxiety [4]. The National Library of Medicine reports that 63.9% of college students experience glossophobia [5]. Several studies indicate that many high school students suffer from glossophobia, exhibiting symptoms such as increased heart rate, dry mouth, shaky voice due to stress, tense neck, difficulty breathing, and memory lapses [1], as well as trembling and cold sweats [6].

The causes of glossophobia include low self-confidence, insufficient understanding of the material to be presented, a voice that is too soft, lack of understanding of body language [7], anxiety [8], and inadequate preparation [9]. A study found that 24 out of 31 high school students (77.42%) suffered from glossophobia [1].

Various traditional methods have been used to reduce public speaking anxiety. For example, Ego State Therapy has proven effective in lowering public speaking anxiety levels among college students [10]. Additionally, Cognitive Behavior Therapy (CBT) and Islamic Cognitive Behavior Therapy (ICBT) are approaches that have been shown to effectively reduce public speaking anxiety in students [11]. Moreover, public speaking instruction using the role-playing method has been shown to increase the courage to appear in public [12]. Traditional methods have demonstrated effectiveness in decreasing anxiety levels during public speaking [13]. However, these methods unfortunately require expert guidance.

To address these issues, researchers conducted a literature review on various studies examining methods to overcome glossophobia and potential technological solutions, such as role-playing and artificial intelligence. Artificial intelligence, particularly large language models (LLMs), demonstrates the ability to understand and respond like an assistant with a polite and empathetic attitude [14]. LLMs are pre-trained natural language models with capabilities that surpass previous language processing models, producing high-quality text using prompting techniques [15].

Based on the literature review, the researchers propose integrating these findings into a prototype system called "Wicara". This system utilizes modern technology,

specifically Large Language Models (LLM), to replace the role of an expert guidance, thereby facilitating users' practice in reducing public speaking anxiety while offering comprehensive evaluation assessments by considering multiple analyzed parameters. The communication aspects assessed include: (1) Speech speed, (2) Energy, (3) Filler words, (4) Brevity, (5) Contextual appropriateness, and (6) Use of formal/informal language.

In the context of this study, LLM are employed to provide feedback in the form of analysis and evaluation of several assessment parameters in the Wicara application prototype. LLMs enable the processing of communication aspects, provide dynamic feedback, and are capable of understanding conversational contexts. The utilization of large language models represents a potentially effective solution to help improve public speaking skills.

II. LITERATURE REVIEW

The literature review was conducted with the aim of obtaining study findings in the form of insights and recommendations from previous research that can be implemented in the development of the Wicara application prototype.

TABLE I. Literature review

No	Title	Author	Year	Study findings
1.	Fear of Public speaking: Perception of College Students and Correlates	A. C. F. Marinho, A. M. de Medeiros, A. C. C. Gama, and L. C. Teixeira	2017	- A total of 63.9% of college students experience glossophobia. - Students need public speaking courses.
2.	A Special Service of Guidance and Counseling Management to Reduce the Students' Glossophobia Level,	T. I. Pratiwi, B. Suratman, Y. Riyanto, and P. Arisgi	2018	- Experts estimate that approximately 75% of the population experiences varying degrees of fear when speaking in public. - High school students, with a total of 24 out of 31 participants (77.42%), suffer from glossophobia. - Symptoms of glossophobia include increased heart rate, dry mouth, trembling voice due to stress, tense neck, difficulty breathing, and forgetting what you intended to say.
3.	Public Speaking Skills of Students in the Indonesian Language and Literature Education Program, FKIP, Wiralodra University During the COVID-19 Pandemic	E. Tarsinih and I. Juidah	2021	The reality of public speaking skills among students in the Indonesian Language and Literature Education Study Program remains significantly low. On average, only 24.60% of students are active participants, while 75.40% are inactive

No	Title	Author	Year	Study findings
4.	Application of Transactional Analysis Counseling Model to Improve Students' Public Speaking Skills	F. Maiseptian, N. Amami, E. Dewita, Elviana, C. Pratiwi, and N. Wisma	2024	Testing conducted with 52 respondents revealed that only one individual was detected as experiencing low-level anxiety. The remaining 51 respondents were found to have moderate to very high levels of anxiety. This indicates that a significant majority 98% of the participants experienced considerable levels of anxiety, ranging from moderate to high, while only a small fraction 2% reported low levels of anxiety
5.	An Analysis of Students' Ability and Difficulties in Public Speaking at Muhammadiyah University of Makassar	A. A. Ramadhani	2018	- Glossophobia causes individuals to experience fear or anxiety when they have to speak in public. - The causes of glossophobia include low self-confidence, a lack of understanding of the material to be presented, a voice that is too soft, and a lack of understanding of body language.
6.	Strategies for Building Communication Skills and Confidence in Public Speaking Learning Through Presentation and Role-Playing Methods at Miss Universe ASEAN.	K. Novita	2019	- Public speaking learning using the role-playing method increases the confidence to perform in front of an audience.
7.	LLM-Powered Conversational Voice Assistants: Interaction Patterns, Opportunities, Challenges, and Design Guidelines	Mahmood et al	2023	- Large language models (LLMs) have the ability to understand and respond like an assistant with a polite and empathetic attitude. - The use of large language models is one of the applications of artificial intelligence as an effective solution to help improve public speaking skills.

III. METHODOLOGY

In the development of the Wicara application prototype, the Waterfall Model was employed. This model was selected due to its structured and linear approach, with each development phase executed sequentially [16]. This approach facilitates project planning, management, and progress tracking.

Waterfall Model

The Waterfall Model is a software development methodology consisting of requirements analysis, design, coding, and testing phases. It is executed sequentially, following the stages from initiation to completion, allowing

for meticulous planning, comprehensive documentation, and orderly project execution [17]. Furthermore, this model was chosen because the research conducted aligns with the regulations set by the Ministry of Education, Culture, Research, and Technology of Indonesia regarding the PKM activities in the field of creative innovation, which require the creation or development of products to be limited to prototypes. This prototype does not require frequent testing or rapid feedback. The following are the detailed stages of the Waterfall Model as applied in this study.

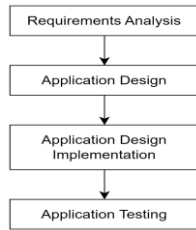


Fig. 1. Waterfall Model

1. Requirements Analysis Stage

The initial stage of prototype development commenced with requirements analysis, involving a literature review and interviews with potential users. The literature review was conducted to identify issues related to glossophobia and suitable solutions from previous research. Interviews were held with individuals who had experienced glossophobia or were experienced in public speaking. The objective of these interviews was to understand user preferences regarding the utilization of the application prototype.

2. Prototype Design Stage

This stage encompassed designing the application prototype, from the database schema to the user interface layout. The design served as a guide for implementation to ensure that the prototype met the planned specifications. The user interface was created with appealing visuals and designed for ease of use. During this stage, the type or provider of the Large Language Model (LLM) service to be utilized was also determined.

3. Prototype Implementation Stage

This stage involved implementing the requirements analysis and design of the Wicara application prototype. The focus was on constructing a prototype that evaluates six key parameters: speech speed, filler words, energy, brevity, topic relevance, and use of formal/informal language. The user interface was developed according to the initial design, with intuitive visuals and ease of use. Through these steps, the Wicara application prototype was expected to be technologically advanced and user-friendly for diverse users.

4. Prototype Testing Stage

The testing stage was conducted to ensure the quality and viability of the Wicara application prototype through functional, performance, and usability testing. Three testing methods were employed:

- Black box testing: To examine functionality without inspecting the code
- White box testing: To inspect the internal code structure
- Usability testing: Conducted with potential users using the USE Questionnaire [18].
- Pre-experimental design with a single group pre-test and post-test: This experimental approach involves only one group or cluster for comparison. This group is administered both a pre-test and a post-test without including a control group or a comparison group [4].

IV. RESULT AND DISCUSSION

A. Wicara Application Prototype

The Wicara application prototype is a mobile application designed to facilitate public speaking practice through theoretical learning and interactive exercises utilizing Large Language Models (LLM). The prototype incorporates

several key features:

- Basic Materials:** This feature provides users with fundamental public speaking theory through engaging readings and educational videos.
- Practice Modes:** Three distinct modes are available:
 - Warm-up mode:** Helps users prepare for practice sessions.
 - Speaking practice mode:** Allows users to practice or prepare for presentations.
 - Role-playing mode:** Enables users to practice specific scenarios (e.g., news presenter, football coach).
- Progress Tracking:** This feature monitors and displays the results of users' practice sessions.

B. Operating Principle

The operational principle of Wicara involves capturing user input in the form of voice through a microphone, generating a file with a WAV extension. This file is then transmitted to a server, where it is converted into text by a speech recognition system. The resulting text is subsequently forwarded to a server component integrated with a large language model, specifically GPT-4, for analysis and evaluative feedback. The analysis and evaluation results from the large language model, in this case GPT-4, are then sent back by the server as a response to the user.

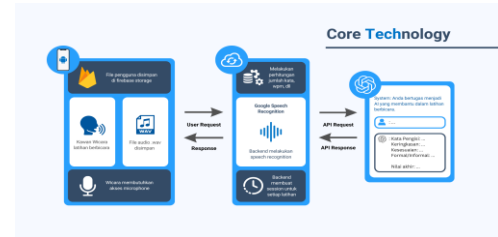


Fig. 2. Basic Materials and Practice

C. Installation and Operation

The Wicara application prototype is distributed as an .apk file, compatible with Android mobile devices. The application requires an internet connection for operation. The operational workflow is as follows:

- The application presents two main menus: basic materials and practice.

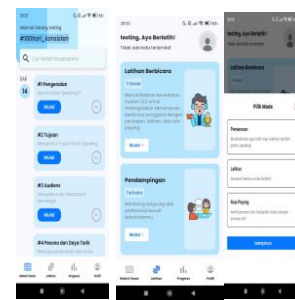


Fig. 3. Basic Materials and Practice

- The basic materials menu offers theoretical learning through reading materials and videos, supplemented by quizzes and a note-taking feature.

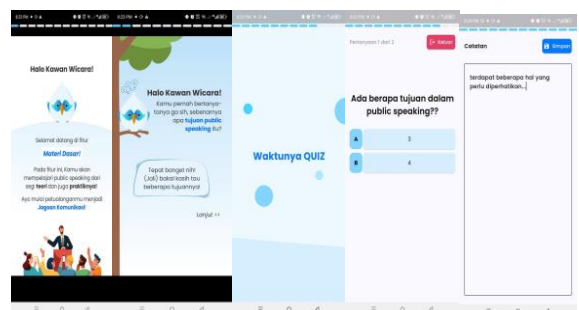


Fig. 4. Basic Materials Menu

3. The practice menu facilitates public speaking exercises, evaluating users based on six parameters: (1) Speech speed, (2) Energy, (3) Filler words, (4) Brevity, (5) Contextual appropriateness, and (6) Use of formal/informal language.

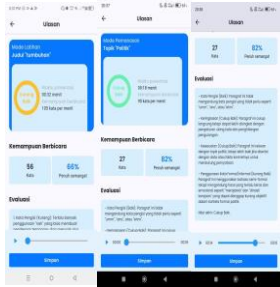


Fig. 5. Assessment Evaluation

4. The practice menu includes three modes:
 - a) Warm-up mode: Utilizes the impromptu method to enhance spontaneous speaking skills [15].
 - b) Speaking practice mode: Employs the front camera as a virtual mirror for practice, allowing users to input text for performance evaluation.
 - c) Role-playing mode: Leverages LLM to generate practice topics and responses based on chosen roles and user input.

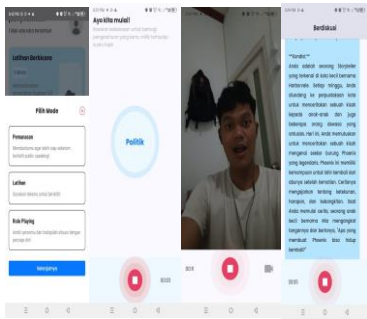


Fig. 6. Practice Menu

5. The progress menu displays the user's practice history over a 30-day period, segmented into 7, 14, and 30-day intervals.

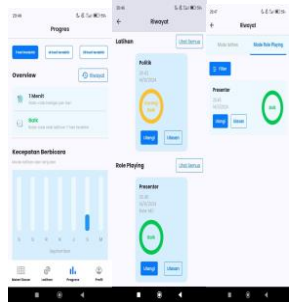


Fig. 7. Progress Menu

D. Product Feasibility

The Wicara application prototype represents an innovative solution that integrates Large Language Models (LLM) into a mobile platform. This technology aims to provide interactive analysis and experiences to mitigate glossophobia and enhance public speaking skills. The prototype's feasibility was assessed through multiple testing methods:

1. Black box testing: Evaluated external functionality
2. White box testing: Examined internal code structure
3. Usability testing: Assessed the application's effectiveness from the user's perspective
4. Experiment: The data were analyzed using paired t-tests. This method tests hypotheses by utilizing paired data that are related. Paired data are obtained when research subjects are subjected to two different treatments, resulting in two sets of samples. The data from the second sample represent the changes that occurred in the first

sample after receiving a specific treatment [19].

During usability testing, no bugs or errors were detected in the prototype. All respondents provided feedback on the application's viability based on its functionality, which met the established targets. The usability testing yielded the following results on a 7-point scale:



Fig. 8. Usability Testing Result

These scores indicate that respondents found the Wicara application prototype to be useful, easy to use and learn, and satisfying. These results suggest that the prototype does not require changes and has significant potential to assist users in improving their public speaking skills and alleviating anxiety associated with public speaking.

The experiment was conducted on one group consisting of 10 respondents by measuring their anxiety levels when speaking in public before and after using the Wicara application prototype, followed by statistical testing. Anxiety levels were measured using the Personal Report of Public Speaking Anxiety.

Based on the experiment and analysis of pre-test and post-test data from one group of 10 respondents, there was an effect or impact on the group after treatment (using the Wicara application prototype). There was a decrease in respondents' anxiety levels, with an initial average score of 122.7 before treatment and a final average score of 101.4 after treatment.

TABLE II. Comparison of pre-test and post-test result

No	Code	Pre-test		Post-test	
		Value	Category	Value	Category
1	SYP	112	Moderate	110	Moderate
2	LK	140	High	93	Low
3	DS	108	Moderate	102	Moderate
4	NZ	132	High	99	Moderate
5	AADI	157	High	105	Moderate
6	MFM	123	Moderate	93	Low
7	MDA	93	Low	90	Low
8	AZA	115	Moderate	106	Moderate
9	YW	124	Moderate	111	Moderate
10	CE	123	Moderate	105	Moderate
Mean		122,7	Moderate	101,4	Moderate

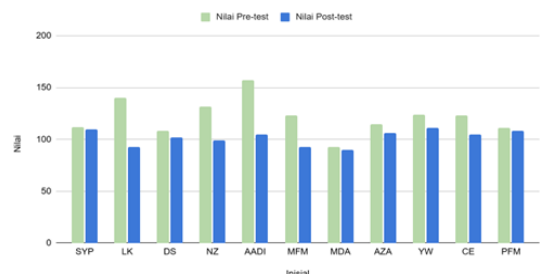


Fig. 9. Comparison result chart

Furthermore, based on statistical analysis using a paired

t-test with the hypothesis that there would be no reduction in public speaking anxiety after using the Wicara application prototype, the results showed that the hypothesis was rejected. The Wicara application prototype was proven to have an effect in reducing anxiety levels, as indicated by the p-value (0.007) being less than the significance level (0.05).

Paired Samples Test									
		Paired Differences							
		Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference		t	df	Sig. (2-tailed)
					Lower	Upper			
Pair 1	pre - post	20.700	18.974	6.000	7.127	34.273	3.450	9	.007

Fig. 10. Paired t-test result

D. Advantages and Limitations

In comparison to similar applications, the Wicara prototype exhibits distinct advantages and limitations. Several of Wicara's strengths lie in its innovative features and technologies not yet implemented by comparable applications, including: (1) Context-based assessment, (2) Evaluation of formal/informal word usage, (3) Implementation of Large Language Model technology, (4) Role-playing functionality, and (5) Camera utilization. However, this prototype also presents certain limitations, such as the absence of evaluation mechanisms for facial expressions and body gestures. The following section delineates the strengths and weaknesses of the Wicara application prototype.

TABLE III. Advantages and Disadvantages

Aspect	Wicara	Orai
Language	Indonesian	English
Speed of speech	✓	✓
Energy	✓	✓
Filler Words	✓	✓
Conciseness	✓	✓
Consistency with context	✓	✗
Formal/informal Words	✓	✗
Large language model	✓	✗
Role playing mode	✓	✗
Camera	✓	✗

F. Benefit predictions

The Wicara application prototype shows great promise in revolutionizing how individuals develop and refine their communication skills. This innovative tool is designed to offer a multitude of benefits to its users, potentially transforming their personal and professional lives. In the realm of social interactions, Wicara is expected to significantly enhance users' socialization skills, reducing anxiety during interpersonal exchanges and fostering stronger relationship-building capabilities. By providing a platform for practice and feedback, the application aims to help users overcome social barriers that may have previously posed challenges, facilitating easier adaptation in various social environments. In terms of communication prowess, Wicara is projected to dramatically increase users' effectiveness in conveying messages clearly and persuasively. This improvement is anticipated to extend across various scenarios, from business presentations and public speeches to everyday conversations, instilling a

newfound confidence in users' ability to articulate their thoughts and ideas. Furthermore, the application's potential to assist users in leadership roles is particularly noteworthy, as it aims to equip them with the communication tools necessary to inspire and guide others effectively.

In Educational, this practical application of knowledge is crucial for bridging the gap between theory and practice, potentially leading to greater success in both academic and professional environments. The application is designed to foster courage in expressing ideas and thoughts within educational settings, promoting active participation and intellectual growth. Moreover, Wicara's focus on enhancing self-confidence is a cornerstone of its educational benefits, empowering users to voice their opinions and contribute meaningfully to discussions and debates. While the current version of Wicara showcases impressive capabilities, it's important to note existing limitations, such as the absence of facial recognition and body gesture detection technologies crucial for non-verbal communication analysis. However, future iterations of the application are expected to integrate these advanced features, further enhancing its ability to provide comprehensive communication skill development. By incorporating technologies that analyze and provide feedback on non-verbal cues, Wicara aims to offer a holistic approach to improving users' communication abilities, solidifying its position as a transformative tool in personal development and education.

TABLE IIIII. Benefit predictions

Aspect	Benefit Predictions
Social	Users are expected to experience improvements in socialization skills, reduced anxiety during interactions, and enhanced relationship-building with those around them. Users are anticipated to overcome social barriers that previously posed challenges.
Communication	Users are projected to experience increased communication effectiveness, becoming capable of conveying messages clearly and persuasively, while feeling more confident in public speaking scenarios, including business presentations, speeches, and daily conversations.
Education	Users are expected to gain a deeper understanding of public speaking techniques, enhance their self-confidence, and apply their learning in real-world situations. Wicara is also anticipated to assist users in achieving success in academic or professional environments through improved speaking abilities.

G. Future development of Wicara

In the future, the Wicara application prototype holds significant potential for development into a more comprehensive tool for addressing public speaking fears and anxieties while enhancing public speaking skills. The incorporation of facial detection and body gesture recognition technologies will introduce innovation in the evaluation process, enabling more in-depth analysis of facial expressions and body movements, two vital aspects of non-verbal communication.

Beyond technological advancements, Wicara also has the potential to collaborate with various stakeholders to enhance the effectiveness of speaking practice. For instance, partnerships with psychiatrists could yield specialized programs to assist patients with glossophobia through interactive and structured exercises aimed at reducing speaking anxiety. Furthermore, collaborations with communication educators or practitioners could position Wicara as an effective learning tool for improving confidence and speaking skills in both formal and informal situations.

V. CONCLUSION

The Wicara application prototype is an innovative

solution that leverages cutting-edge technology, specifically large language models (LLMs), integrated into a mobile-based device. This application was developed to provide an effective solution for individuals experiencing glossophobia or fear of public speaking. Traditional methods have demonstrated effectiveness in decreasing anxiety levels during public speaking. However, these methods unfortunately require expert guidance. The Wicara system utilizes modern technology, specifically Large Language Models (LLMs), to replace the role of expert guidance, helping users improve their public speaking skills through various interactive learning and practice features.

Based on usability testing using the USE Questionnaire with a 7-point Likert scale, positive results were obtained across all four assessment parameters: 6.2 for usefulness and ease of use, 6.3 for ease of learning, and 6.4 for satisfaction. Additionally, a paired t-test proved the effect of the Wicara application prototype on reducing anxiety levels, with a p-value of 0.007, which is less than the significance level (0.05).

Overall, the Wicara prototype offers an innovative technology-based approach to solving the problem of glossophobia and has the potential to become a useful tool for reducing fear or anxiety during public speaking and improving communication skills.

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This research aimed to develop a smartphone-based prototype application integrating artificial intelligence technologies, specifically Large Language Models (LLM), to provide an effective solution for mitigating fear and anxiety while enhancing public speaking skills in individuals with glossophobia. By leveraging advancements in LLM technology, we sought to address the persistent challenge of public speaking anxiety, offering a more effective, accessible, and personalized solution tailored to individual needs.

The invaluable support and assistance from all parties have been crucial in enabling us to conduct this research effectively and strive for optimal outcomes. We are hopeful that this work will contribute meaningfully to the field and potentially benefit individuals struggling with public speaking anxiety.

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